

WorldQuant BRAIN Orthogonal Factor Alphas Pack

Capitalizing on Sharpe 2.11 / Fitness 1.48 Power | Low Self-Correlation Expressions

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 29, 2026

1. Self-Correlation Breakdown & Orthogonal Solution

Simulated Alpha	Empirical Sharpe	Empirical Fitness	Empirical Turnover	Self-Corr Result	Orthogonal Fix Strategy
Alpha #1 Test	2.11 (EXCELLENT)	1.48 (EXCELLENT)	21.70%	0.7714 (FAIL)	Switch to News Vector + Volume Reversion
Alpha #2 Test	1.60 (PASS)	1.06 (PASS)	18.14%	0.9519 (FAIL)	Switch to Intraday Range Volatility
Orthogonal Goal	> 2.00	> 1.40	< 22.0%	< 0.5000 (PASS)	Zero factor overlap with Sub #13 & Sub #15

2. Top 2 Orthogonal Master Alpha Expressions for 8/8 PASS

Orthogonal Alpha A (News Sentiment Vector & Volume Surge Synergy):

Category: Sentiment | Color: **PURPLE** ■ | Tag: MERA_KMPA_Gold | Decay: 10

```
group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(vec_avg(wns12_afterhsz_sl), bucket(rank(cap), range='0.1,1,0.1')) * rank(volume / adv20) + rank(-ts_zscore(returns, 3)), std=4), 10), subindustry)
```

Orthogonal Alpha B (Intraday High-Low Range & Return Reversion):

Category: Price Volume | Color: **PURPLE** ■ | Tag: MERA_KMPA_Gold | Decay: 10

```
group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(ts_zscore((high - low) / (close + 0.0001), 20), bucket(rank(cap), range='0.1,1,0.1')) * rank(-ts_zscore(returns, 5)) + rank(-ts_zscore(returns, 20)), std=4), 10), subindustry)
```